Introduction to Satellite Systems

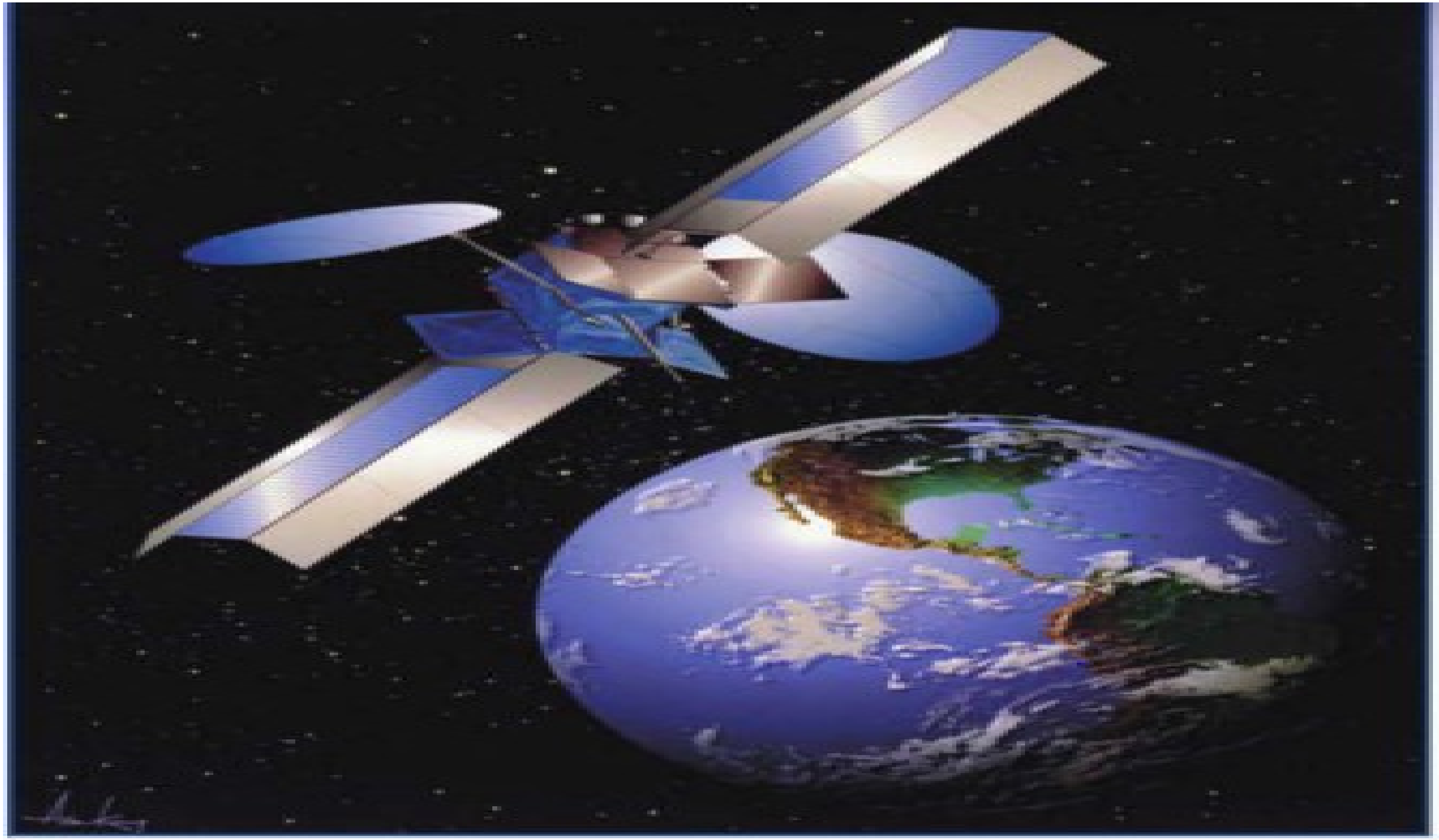
A satellite is an artificial object which is placed intentionally into an orbit of any natural satellite. Satellites are used for many purposes i.e., weather forecasting, digital transmission, scientific research and development etc.

- In a communication context, a satellite is a specialized wireless transmitter/receiver that is launched by a rocket and placed in orbit around the earth.
- A satellite can be natural, like the moon, or artificial (human made). So, we can say that a satellite is an object that moves in a curved path around a planet.

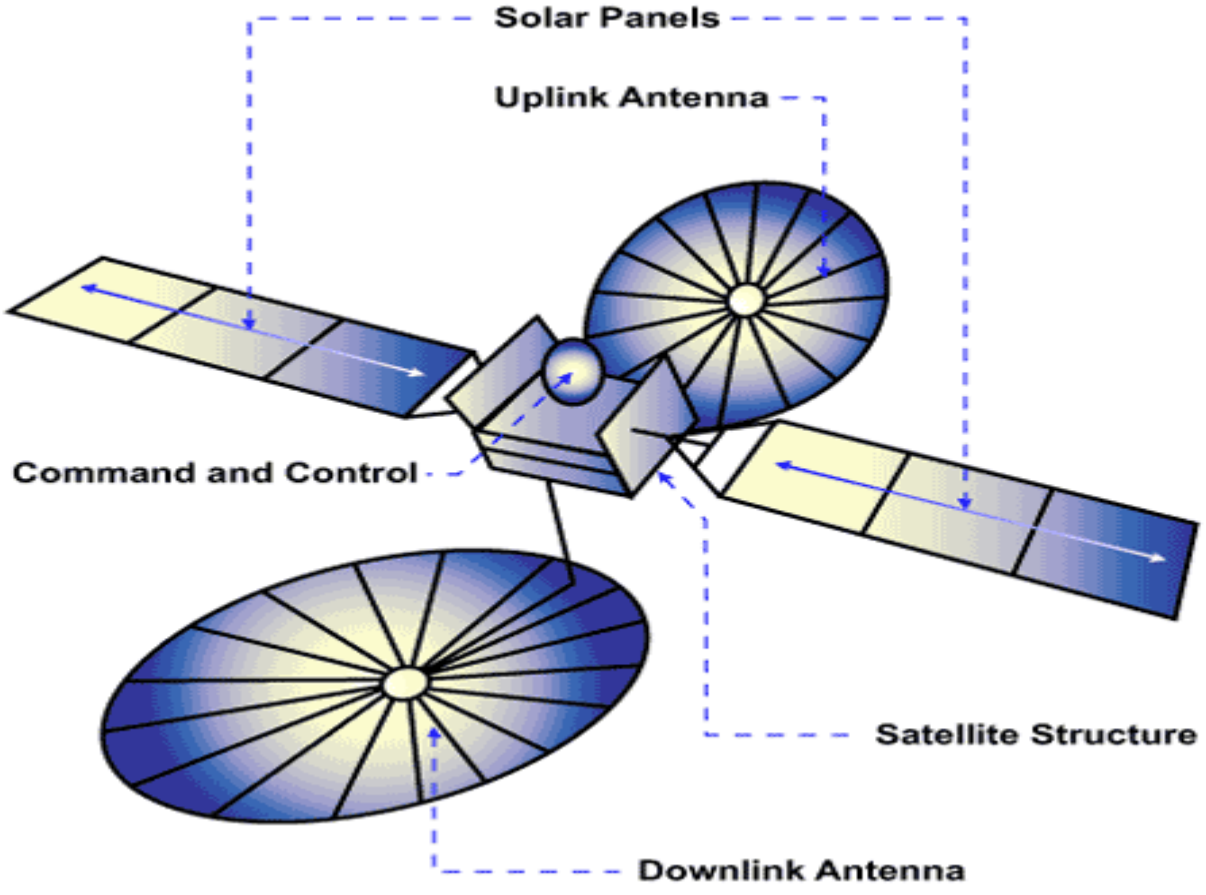
SATELITE AND MOBLIE COMMUNICATION

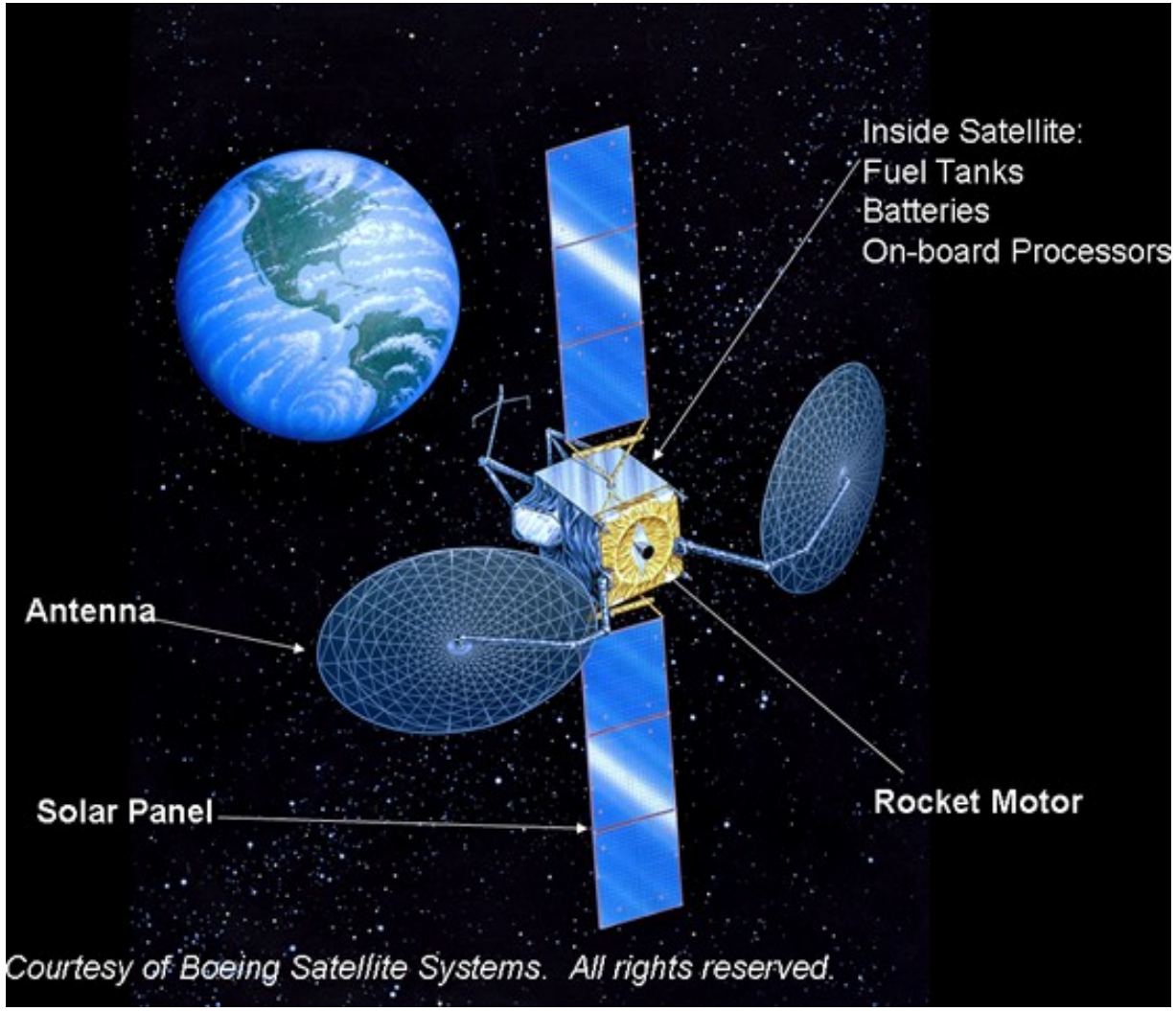
INTRODUCTION

Satellite communication began after the Second World War. Scientists knew that it was possible to build rockets that would carry radio transmitters into space. Ln 1945, Arthur C. Clarke published his essay on 'Extra Terrestrial Relays'. But it was not until 1957, in the middle of the cold war, that the sudden launching of the first satellite SPUTNIK by the Soviet Union shocked the Western world. SPUTNIK is not at all comparable to a satellite today, it was basically a small sender transmitting a periodic 'beep'. But this was enough for the US to put all its effort into developing its first satellite. Only three years later, in 1960, the first reflecting communication satellite ECHO was in space. ECHO was basically a mirror in the sky enabling communication by reflecting signals. Three years further on, the first geostationary (or geosynchronous) satellite SYNCOM followed. Even today, geostationary satellites are the backbone of news broadcasting in the sky. Their great advantage, is their fixed position in the sky.



PARTS OF SATELLITE





Internal computer
Attitude control
Grapple fixture
Body
Orbit
Power source
Communications

- Our final category of wave propagation is satellite communications (SATCOM). Communications via satellite is possible because of the placement of the satellites in geostationary orbit (sometimes called geosynchronous or synchronous orbit).
- This means that the satellite is located at a fixed point approximately 22,300 miles in altitude above the equator. At this altitude, the gravitational pulls of the earth, the sun, and the moon work together, along with the centrifugal force caused by the satellite's rotation around the earth to keep the satellite at a fixed location above the earth. Of course, the satellite does drift (it moves in a figure-eight pattern) and must be periodically repositioned by on-board power thrusters to maintain the optimum location. But for us on earth, the satellite appears to be stationary.

The uplink and downlink are called an earth station (ground base station), which will be typically transmitting and receiving data, video and/or audio, or it can be a receiving only site.

Included in the earth station are items such as the exciter, the high-power TWTAs **(traveling wave tube** amplifiers-also called high power amplifiers [HPAs]), a parabolic shaped reflector that is pointing at the satellites parked in geostationary orbit, and a receiver.
The satellites require a payload of antennas, transponders, and attitude controls for maintaining their location in geostationary orbit.

The transponder is an electronic system performing reception, frequency translation, and retransmission. The attitude controls are used for orbital corrections (station keeping) on the satellite. These corrections are made approximately every 2 to 6 weeks. Geostationary satellites are parked over the equator (latitude = 0°) in orbit at a fixed longitude

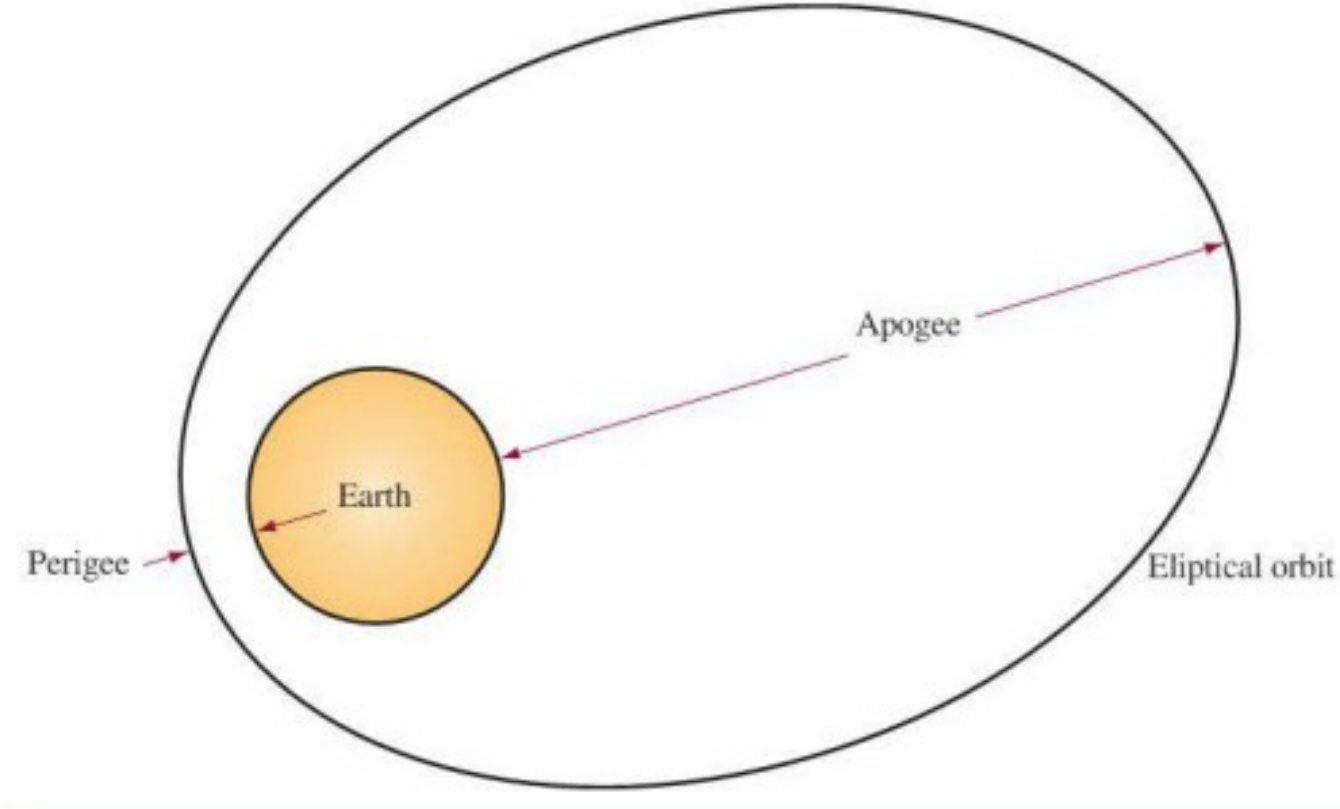
Uplink (transmitter) •
Orbiti ng satellite •
Downlink (receiver)

- The most common frequency bands used in satellite communications are C-band and Ku-band. Table l provides a list of frequencies commonly used in satellite communications for many different applications such as data, voice, entertainment, news, military, international broadcast, among others. Table l does not list all of the satellite frequencies used in the United States but provides th

Table 1 Satellite Frequency Bands

Band	Uplink (GHz)	Downlink (Ghz)
L	1–2	Various
S	1.7–3	Various
C	5.9–6.4	3.7–4.2
X	7.9–8.4	7.25–7.75
Ku	14–14.5	11.7–12.2
Ka	27–30	17–20
	30–31	20–21

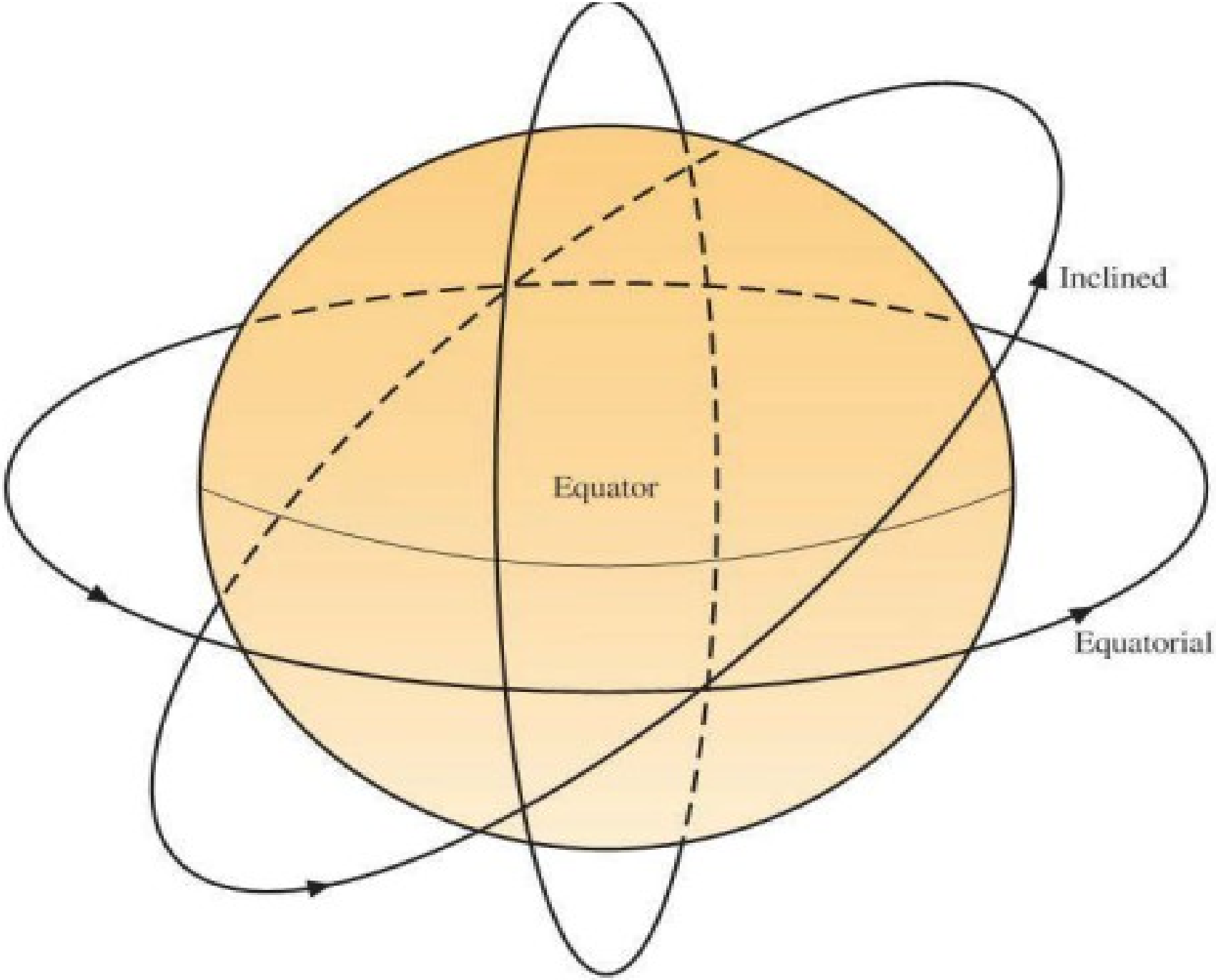
- ORBITAL PATTERNS
- The orbital patterns of satellites are elliptical. For each orbit there is a perigee (closest distance of the orbit to earth) and an apogee (farthest distance of the orbit from earth).
- This is shown in Figure below Geostationary orbits use an equatorial orbit. Other possible satellite orbits are polar and inclined orbits. These orbits are shown in Figure 19. The geostationary satellites use an equatorial orbit .



It is interesting to note that three geostationary orbits, parked 120° apart, can cover the entire earth surface except for the polar regions above latitudes 76 N and 76 S.

This assumes a minimum elevation of 5° for the receiving antennas. A satellite in a polar orbit can cover 100 percent of the earth's surface.

This is possible because the earth is rotating as the satellite travels around the North Pole and the South Pole. Every location on earth is visible to the satellite twice a day. to reach the extreme northern and southern latitudes.


Inclined
Equator
Equatorial

The problem with inclined orbits is that the receiving station must track the satellite. An example of a highly inclined orbit is the Russian Molniya, which has a 63° inclination angle and an orbital period of 12 hours.

The apogee region is when the satellite is above the Northern Hemisphere. At this point, the satellite is easiest to track.

The satellite is visible to earth stations from 4.5 to 10.5 hours each day.

- The azimuth and elevation angles for the earth station antenna must be calculated so that
- the correct satellite can be seen. This is called the look angle.
- **The azimuth** is the horizontal pointing angle of the earth station antenna. The elevation is the angle at which we look up into the sky to see the satellite.
- To calculate the azimuth and elevation of a ground station antenna requires that the ground station latitude and longitude as well as
- the longitude of the satellite are known.

- The latitude and longitude of the earth station can be obtained from U.S. Geological Survey maps or using a Global Positioning Satellite (GPS) receiver.
- Once the actual location is known, the elevation angle of the earth station antenna can be calculated by using either an online calculator such as the Java Script program available at http://web.nmsu.edu/-jbeasley/Satellite/
- Please note, this calculator was developed for an earth station site in the west longitude and north latitude

$$\tan(E) = \frac{\cos(G)\cos(L) - .1512}{\sqrt{1 - \cos^2(G)\cos^2(L)}}$$

where E = elevation in degrees S = satellite longitude in degrees N = site longitude in degrees G = S - N in degrees L = site latitude in degrees

$$A = 180 + \arctan\left(\frac{\tan(G)}{\sin(L)}\right)$$

where A = azimuth of the antenna in degrees S = satellite longitude in degrees N = site longitude in degrees L = site latitude in degrees G=S-N

Problem: Calculate the azimuth and elevation angles for an earth station (ground station) antenna given a satellite longitude of 83° west, a site longitude of 90° west, and a site latitude of 35° north.

Using Equation 8, the azimuth is equal to

$$A = 180 + \arctan\left(\frac{\tan(-7)}{\sin(35)}\right)$$

$$A = 180 + \arctan\left(\frac{-.128}{.5736}\right)$$

$$A = 168°$$

The elevation angle is calculated using Equation 9

$$\tan(E) = \frac{\cos(-7)\cos(35) - .1512}{\sqrt{1 - \cos^2(-7)\cos^2(35)}}$$

$$\tan(E) = \frac{.661846}{.582199} = 1.1368$$

$$\therefore E = \arctan(1.1368) = 48.663°$$

- GLOBAL POSITIONING SATELLITE (GPS)

The GPS is another application made possible by satellite technology. It provides pinpoint geographic location information.

GPS was originally used by the government and law enforcement agencies, but the availability of low-cost handheld receivers has enabled personal use. You can now obtain your exact location when traveling by car or boat or when hiking.

The GPS satellites transmit position data signals, and a GPS receiver processes and computes the time to receive each one. Doing this from four different satellites allows the receiver to determine your exact latitude and longitude.

GPS currently uses a constellation of 28 satellites orbiting above the earth at a distance of 10,900 miles. The GPS satellites complete an orbit about every 12 hours.

- The satellites transmit two signals, a course acquisition (C/ A) signal transmitted on 1575.42 MHz, which is available for civilian use, and a precision code (P-code), transmitted on 1227.6 MHz and 1575.42 MHz, which are for military use only.
- GPS receivers measure the time it takes for the satellite signals to travel from the satellites to the receiver; from this information, the receiver can fix our position (i.e., locate where we are). It takes three satellites to fix our position in terms of latitude and longitude, whereas it takes four satellites to determine three-dimensional information: latitude, longitude, and elevation.
- Civilian receivers can have a position accuracy of about 2 meters, but this distance can vary. The position accuracy can be improved by using a technique called differential GPS. Receiver accuracy is improved by using a ground receiver at a known location to provide corrections to the satel1ite civilian signal error. With differential GPS, the accuracy of a GPS receiver can be improved to about l cm.

Emerging technologies in satellite communication

Emerging technologies in satellite communication are transforming the landscape of global connectivity, enhancing capabilities, and expanding applications. Here are the key developments shaping the future of satellite communications:

1. High Throughput Satellites (HTS)

HTS technology significantly increases the capacity and efficiency of satellites by utilizing advanced spot beam technology. This allows for higher data rates and improved connections, which is particularly beneficial for broadband internet services in both urban and remote areas.

2. Low Earth Orbit (LEO) Satellites

The deployment of LEO satellites is revolutionizing satellite communication by reducing latency and enhancing data transmission rates. These satellites operate closer to Earth, making them ideal for applications like internet services and IoT connectivity. Major players like SpaceX (Starlink) and Amazon (Project Kuiper) are leading efforts to establish large constellations of LEO satellites, promising global coverage.

3. Inter-Satellite Links (ISLs)

- ISLs enable satellites within a constellation to communicate directly with each other, bypassing ground stations. This technology enhances data relay efficiency and reduces latency, facilitating real-time communication across vast distances.

. Software-Defined Satellites (SDS)

SDS technology allows for the reconfiguration of onboard software and hardware while in orbit. This flexibility enables satellites to adapt to changing communication needs and mission requirements dynamically, optimizing resource use and enhancing operational efficiency.

5. Quantum Communication

Research into quantum communication aims to create ultra-secure networks using quantum entanglement principles. Quantum key distribution (QKD) is a significant application that could revolutionize data security in satellite communications, making them impervious to eavesdropping.

6. Integration with 5G Networks

Satellites are increasingly being integrated into 5G ecosystems to provide seamless connectivity, especially in underserved areas. This hybrid approach enhances network resilience and ensures comprehensive coverage across regions lacking terrestrial infrastructure.

7. Artificial Intelligence (AI) and Machine Learning (ML)

- AI and ML technologies are being incorporated into satellite systems to improve data processing, optimize resource allocation, and enable autonomous operations. These advancements enhance capabilities in areas like Earth observation, disaster management, and operational efficiency.

Advanced Propulsion Systems

New propulsion technologies are extending the operational life of satellites and allowing for more flexible orbit adjustments. These advancements improve manoeuvrability and reduce costs associated with satellite deployment and maintenance.

9. Satellite Internet of Things (IoT)

The Satellite IoT trend is enabling unprecedented connectivity across various industries, facilitating real-time data collection and analysis from remote sensors and devices. This integration supports enhanced monitoring capabilities in sectors such as agriculture, logistics, and environmental management.

10. Space Debris Mitigation Technologies

As the number of satellites increases, managing space debris has become critical. Innovative solutions like active debris removal systems and advanced collision avoidance technologies are being developed to ensure the sustainability of satellite operations

- **Weather forecasting**: Several satellites deliver pictures of the earth using, e.g., infrared or visible light. Without the help of satellites, the forecasting of hurricanes would be impossible.
- **Radio and TV broadcast satellites**: Hundreds of radio and TV programs are available via' satellite. This technology competes with cable in many places, as it is cheaper to install and, in most cases, no extra fees have to be paid for this service. Today's satellite dishes have diameters of 30-40 cm in central Europe.
- **Military satellites**: One of the earliest applications of satellites was their use for carrying out espionage. Many communication links are managed via satellite because they are much safer from attack by enemies.
- **Satellites for navigation**: Even though it was only used for military purposes in the beginning, the global positioning system (GPS) is nowadays well-known and available for everyone, The system allows for precise localization worldwide, and with some additional techniques, the precision is in the range of some metres Almost all ships and aircraft rely on GPS as an addition to traditional navigation systems.
- **Global telephone backbones**: One of the first applications of satellites for communication was the establishment of international telephone backbones. Instead of using cables it was sometimes faster to launch a new satellite (aka "big cable in the sky). However, while some applications still use them, these, satellites are increasingly being replaced by fiber optical cables crossing the oceans.

CONT. APPLICATIONS

Remote Area Connectivity

Satellites provide essential communication services to isolated regions lacking infrastructure. This includes connecting communities to global networks for education, healthcare, and economic opportunities.

Advanced Applications

- Emerging technologies such as high-throughput satellites (HTS) and integration with 5G networks are expanding the capabilities of satellite communications. These advancements aim to improve bandwidth availability and reduce latency for various applications. In conclusion, satellite communication is integral to modern life, supporting a wide array of applications that enhance connectivity, safety, and knowledge across the globe. As technology evolves, the potential for satellite communications will continue to grow, paving the way for new innovations and services.

- **Telemetry, Tracking, and Command Subsystem (TT&C)**
- **Function**: TT&C monitors the satellite's health and status while also controlling its operations.
- **Components:** This subsystem collects data from various sensors, transmits it back to ground control, and receives commands from operators.
- **3. Power Subsystem**
- **Function**: It generates, stores, and distributes electrical power to all satellite components.
- **Components**: Typically includes solar panels for energy collection, batteries for storage, and power regulation systems to ensure stable voltage levels.
- **4. Communication Subsystem**
- **Function**: This is the core subsystem for transmitting and receiving signals between the satellite and ground stations.
- **Components**: It comprises antennas, transponders (which amplify and relay signals), and signal processing units that handle modulation, demodulation, encoding, and error correction.
- **5. Payload SubsystemD**
- **Function:** Houses the instruments specific to the satellite's mission (e.g., cameras for Earth observation or transponders for communication).
- **Importance:** The design of this subsystem directly impacts the satellite's mission capabilities.

SATELLITE SUBSYSTEM

Earth Segment Subsystems

The earth segment encompasses all components located on the ground that interact with the satellite. Key subsystems include:

1. Ground Stations

i. **Function**: These facilities transmit signals to satellites and receive data back from them.

ii. **Components**: Ground stations typically include large antennas (like parabolic dishes), transmission equipment, and data processing units.

2. User Terminals

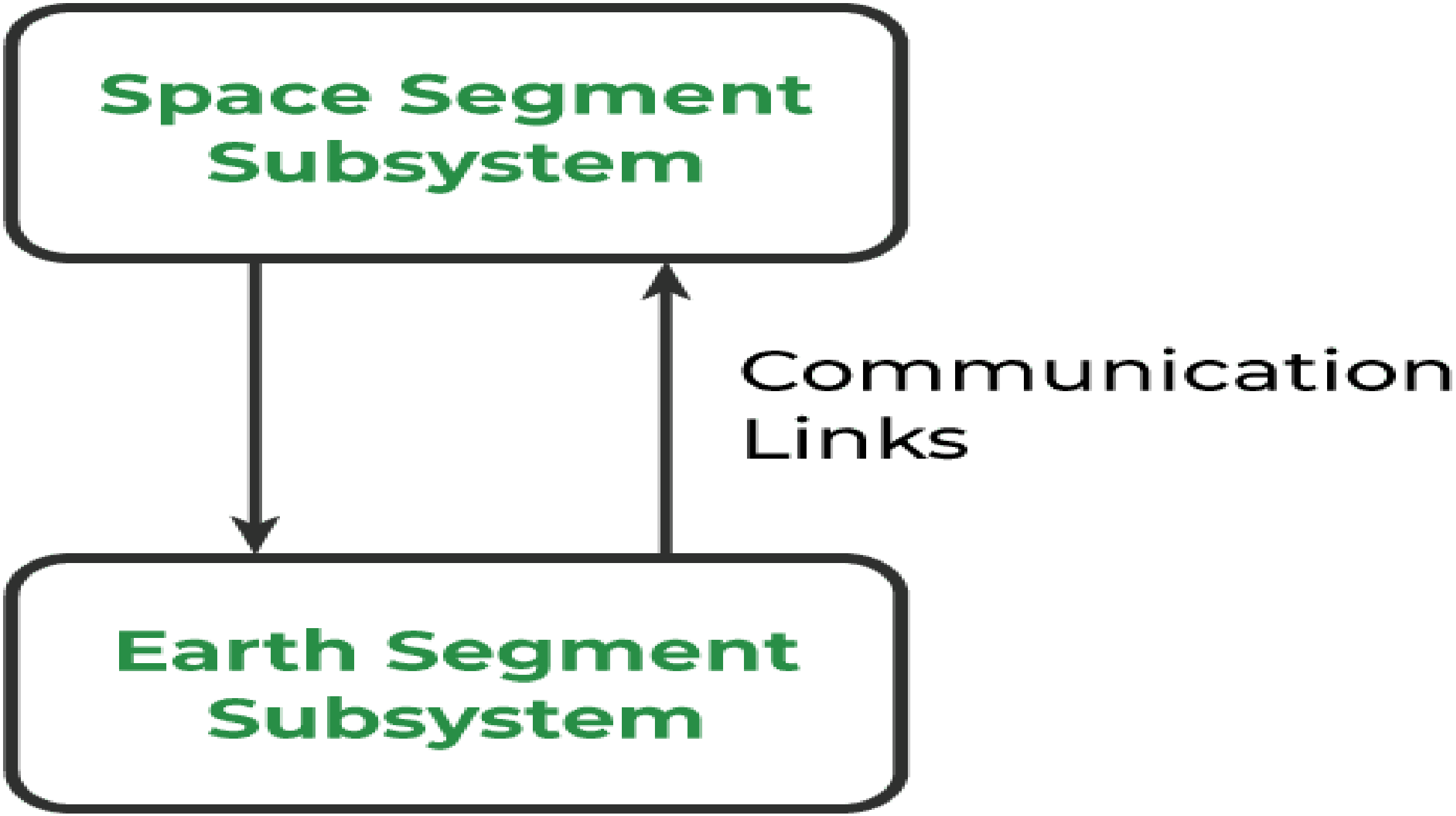
iii. **Function**: These are smaller devices used by end-users to access satellite services.

iv. **Examples**: Satellite phones, broadband terminals, and other communication devices that connect directly with satellites.

3. Data Processing Centres

v. **Function**: These centres analyse and interpret data received from satellites.

Importance: They play a crucial role in applications like weather forecasting, disaster management, and scientific research by processing satellite data into usable information


Space Segment Subsystem
Communication Links
Earth Segment Subsystem

Discuss earth station

Earth stations are integral components of satellite communication systems, serving as the ground-based facilities that facilitate communication with satellites in orbit. They play a crucial role in transmitting and receiving signals, ensuring reliable connectivity for various applications such as telecommunications, broadcasting, and remote sensing.

Types of Earth Stations

Earth stations can be categorized based on their capabilities:

1. Transmit-Only Stations

These stations are designed solely to send signals to satellites without receiving any data back.

2. Receive-Only Stations

These stations only capture signals from satellites and do not transmit data.

3. Transmit-Receive Stations

- These are versatile stations capable of both sending and receiving data, facilitating two-way communication.

Design Considerations

Designing an earth station involves several factors:

i. **Type of Service**: Different requirements exist for Fixed Satellite Service (FSS), Broadcast Satellite Service (BSS), and Mobile Satellite Service (MSS).

ii. **Communication Requirements**: The design must accommodate specific needs such as telephony, data transmission, or television broadcasting.

iii. **Performance Metrics**: Key parameters include Effective Isotropic Radiated Power (EIRP), receiver sensitivity, noise temperature, and antenna pointing accuracy.

Terminals and devices in satellite communication.

In satellite communication, terminals and devices play a crucial role in enabling effective transmission and reception of signals between satellites and ground stations. These components are essential for various applications, including telecommunications, broadcasting, and data services. Here is a detailed look at the key types of terminals and devices used in satellite communication

1. Ground Terminals

Ground terminals are facilities equipped to communicate with satellites. They can be categorized based on their functionality

a. Fixed Earth Stations

i. **Description**: These are stationary installations that maintain a constant connection with satellites, typically used for broadcasting or telecommunications.

ii. **Components**: Fixed earth stations generally include large parabolic antennas, transmitters, receivers, and signal processing equipment.

b. Mobile Earth Stations

i. **Description**: Portable units that can be deployed in various locations, often used in maritime or emergency scenarios.

ii. **Features**: These stations are designed for mobility and may include compact antennas and integrated communication equipment.

c. User Terminals

iii. **Description**: Devices used by end-users to access satellite services directly.

iv. **Examples**: Satellite phones, broadband modems, and VSAT (Very Small Aperture Terminal) systems that provide internet access in remote areas.

2. Antennas

- Antennas are critical devices in satellite communication, responsible for transmitting and receiving electromagnetic waves.

a. Parabolic Dish Antennas

i. **Function**: Used primarily in fixed earth stations to focus signals on the satellite.

ii. **Characteristics**: These antennas must be accurately aligned with the satellite to maintain a strong connection.

b. Phased Array Antennas

iii. **Function**: These antennas can electronically steer their beam direction without physical movement.

iv. **Advantages**: They offer rapid tracking capabilities for satellites in low Earth orbit (LEO), making them suitable for mobile applications.

3. Transmitters and Receivers

Transmitters send signals to satellites, while receivers capture signals from them.

a. Block Upconverters (BUC)

v. **Function**: Converts lower frequency signals to higher frequencies for uplink transmission.

vi. **Importance**: BUCs are essential for ensuring that signals can be effectively transmitted over long distances to satellites.

b. **Low Noise Block Converters (LNB)**

I. **Function**: Used to receive downlink signals from satellites and convert them to a lower frequency for processing.

II. **Role**: LNBs improve signal quality by minimizing noise during reception.

4. Signal Processing Equipment

This category includes devices that enhance the quality and reliability of satellite communications.

a. Satellite Modems

i. **Function**: Modulate and demodulate signals for transmission over satellite links.

ii. **Capabilities**: Advanced modems can implement error correction techniques to ensure data integrity even under adverse conditions.

b. Amplifiers

i. **Types**: This includes power amplifiers and solid-state power amplifiers (SSPA).

ii. **Purpose**: Amplifiers boost signal strength before transmission or after reception to ensure clear communication.

5. Additional Devices

- Several other devices support the overall functionality of satellite communication systems:

a. Transceivers

i. Description: Integrated devices that combine both transmitter and receiver functionalities.

ii. Use Case: Commonly found in user terminals for streamlined communication.

b. Splitters and Switches

iii. Function: Splitters divide a signal into multiple paths, while switches route signals between different paths or devices.

iv. Importance: These devices enhance flexibility in managing signal distribution within the network.

c. Frequency Converters

v. Function: Convert microwave signals into lower or intermediate frequency ranges for further processing.

vi. Components: Typically consist of mixers, filters, and amplifiers to ensure proper signal conversion.

Transponder.

A transponder in satellite communication is a critical component that facilitates the transmission and reception of signals between ground stations and satellites. It essentially acts as a relay, receiving signals from the Earth, processing them, and then retransmitting them back to the ground or to other satellites. Here is a detailed overview of transponders, their functions, types, and significance in satellite communication.

Components of transponder:

i. **Input Band-Limiting Device**: Filters incoming signals to limit bandwidth.

ii. **Low Noise Amplifier (LNA)**: Amplifies weak signals received from Earth stations.

iii. **Frequency Translator**: Converts the frequency of the received signal to the frequency required for transmission.

iv. **Output Band-Pass Filter**: Ensures that only the desired frequencies are transmitted.

v. **Power Amplifier**: Boosts the strength of the downlink signal before it is sent back to Earth.

Functions of a Transponder

i. **Signal Reception**: Captures signals from ground stations.

ii. **Amplification**: Strengthens weak incoming signals.

iii. **Frequency Translation**: Changes the frequency of the signal for retransmission.

iv. **Signal Processing**: May include demodulation and re-modulation, especially in regenerative transponders.

Satellite power systems.

- Satellite power systems are essential for the operation of satellites, providing the necessary electrical energy to support various subsystems and payloads. These systems are designed to generate, store, and distribute power efficiently, ensuring that satellites can function effectively throughout their missions. Here’s a detailed discussion of the components, functions, and technologies involved in satellite power systems.

Key Components of Satellite Power Systems

1. Solar Arrays

i. **Function**: Solar arrays are the primary source of power for most satellites. They convert solar energy into electrical power using photovoltaic cells.

ii. **Design**: Modern solar arrays can achieve efficiencies of around 30% and are often composed of multiple solar cells connected in series or parallel to generate sufficient power. The size of the solar array is crucial; larger arrays capture more sunlight, which is especially important for high-demand missions.

2. Batteries

iii **Function**: Batteries store energy generated by solar arrays for use during periods when the satellite is in the Earth's shadow or when solar power is insufficient.

iv. **Types**: Lithium-ion batteries are commonly used due to their high energy density and efficiency. These batteries ensure a continuous power supply, particularly critical during eclipse periods.

. **Power Conditioning Unit (PCU)**

i. **Function**: The PCU manages the distribution of electrical power to various subsystems within the satellite. It regulates voltage levels and ensures that each subsystem receives the appropriate amount of power.

ii. **Features**: The PCU includes components like voltage regulators, switches, and fuses to protect against electrical failures and manage power flow efficiently.

4. Power Distribution System

iii. **Function**: This system distributes electrical power from the solar arrays and batteries to different satellite subsystems based on their operational needs.

iv. **Importance**: It ensures that critical systems receive power while managing load demands and preventing overloading or underutilization.

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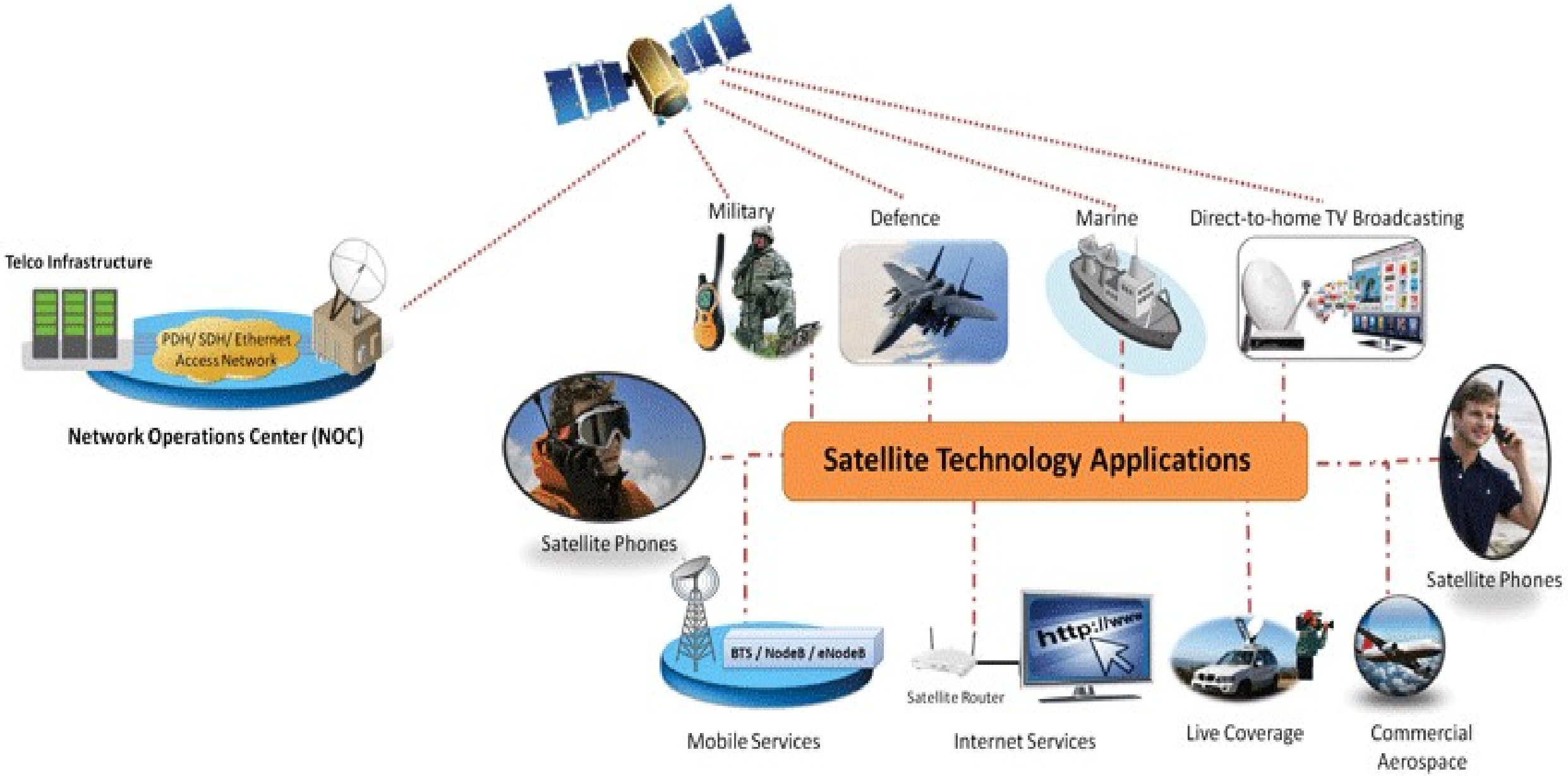
APPLICATION OF SATELLITE



Describe Satellite subsystems.

Satellite communication systems are complex structures composed of various subsystems that work together to ensure effective operation. These subsystems can be broadly categorized into two segments: **space segment subsystems** and **earth segment subsystems**. Below is an overview of these subsystems and their functions.

Space Segment Subsystems

The space segment refers to the components located in space, primarily the satellite itself. Key subsystems within the space segment include:

1. Attitude and Orbit Control Subsystem (AOCS)

i. **Function**: This subsystem is responsible for maintaining the satellite's orientation (attitude) and its position in orbit.

ii. **Components:** It includes sensors like star trackers and gyroscopes, as well as actuators such as reaction wheels and magnetic torquers that adjust the satellite's orientation based on feedback from the sensors.

TYPES OF SATELLITES

GEOSTATIONARY (OR GEOSYNCHRONOUS) EARTH ORBIT (GEO)

•GEO satellites have a distance of almost 36,000 km to the earth. Examples are almost all TV and radio broadcast satellites, many weather satellites and satellites operating as backbones for the telephone network.

•If a satellite should appear fixed in the sky, it requires a period of 24 hours. Using the equation for the distance between earth and satellite r = (g-R/(2-T))s and the period of 24 hours f= 1/24h, the resulting distance is 35,786 km. The or

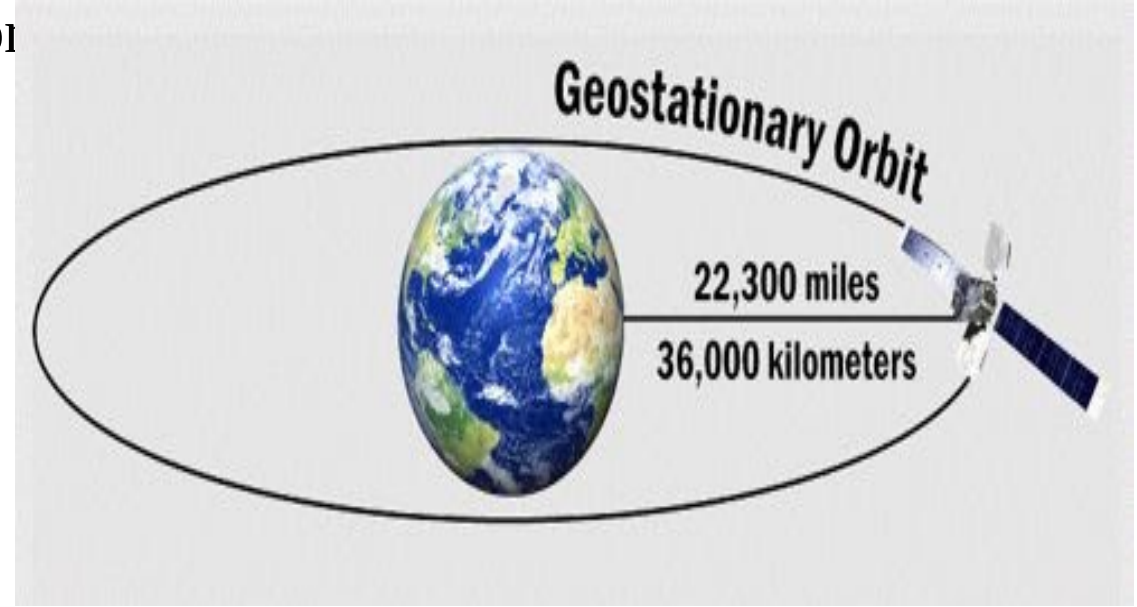
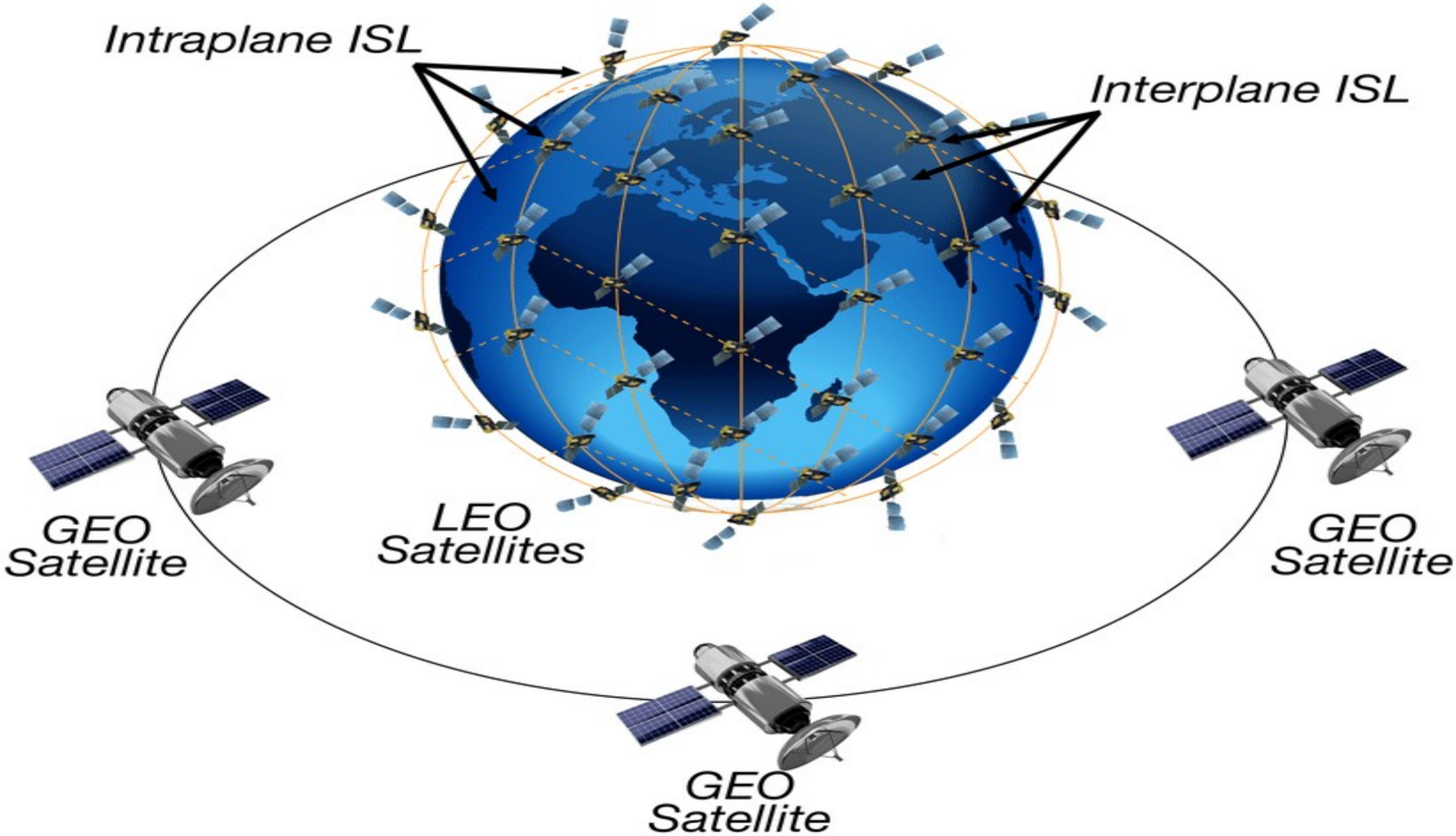


Fig. 1

Geostationary
Orbit Tutorial


Intraplane ISL
Interplane ISL
GEO
Satellite
LEO
Satellites
GEO
Satellite
GEO
Satellite

ADVANTAGES OF GEO

Three GEO satellites are enough for a complete coverage of almost any spot on earth. Senders and receivers can use fixed antenna positions, no adjusting is needed. GEOs are ideal for TV and radio broadcasting. Lifetime expectations for GEOs are rather high, at about 15 years. GEOs typically do not need a handover due to the large footprint. GEOs do not exhibit any Doppler shift because the relative movement is zero.

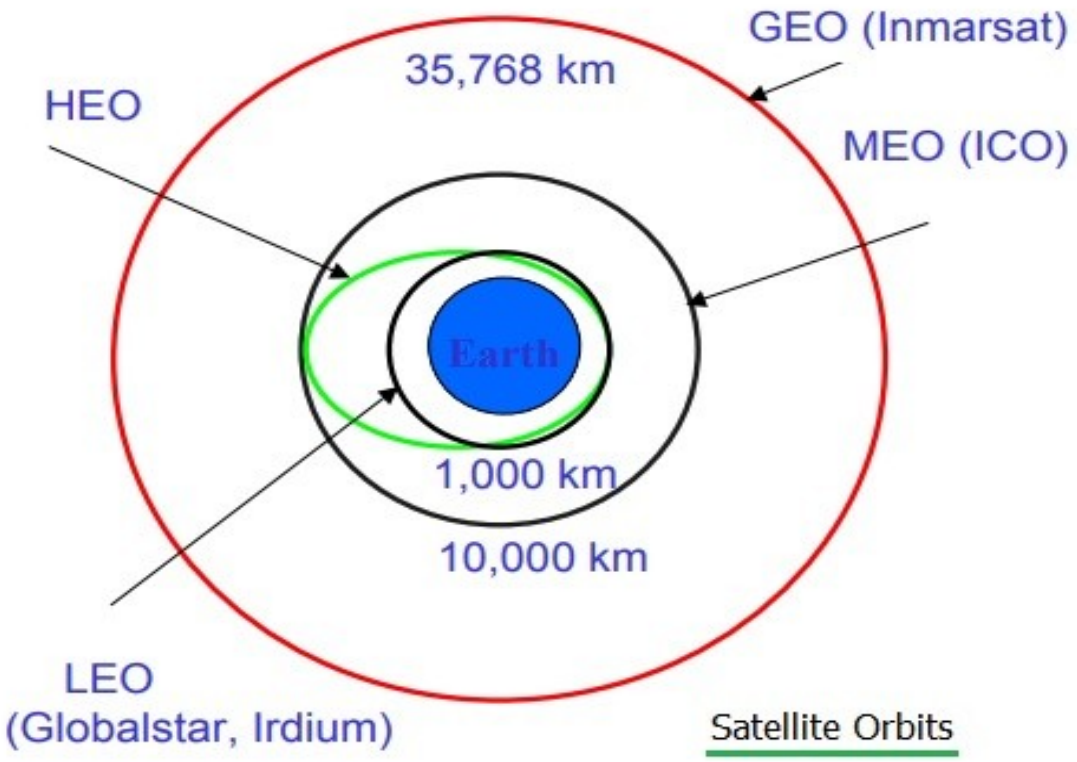
DISADVANTAGES OF GEO

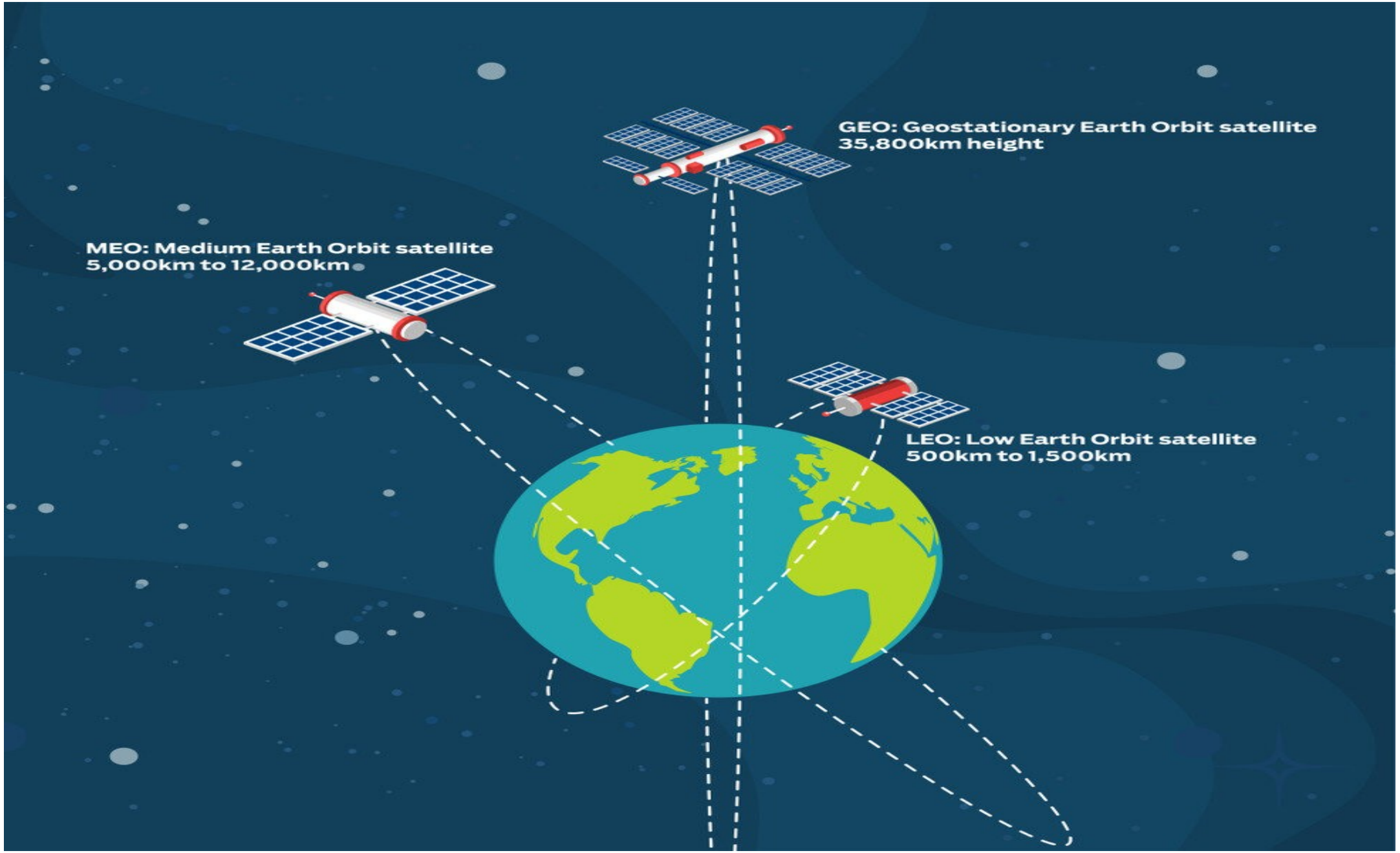
Northern or southern regions of the earth have more problems receiving these satellites due to the low elevation above a latitude of 60, i.e., larger antennas are needed in this case. Shading of the signals in cities due to high buildings and the low elevation further away from the equator limit transmission quality. The transmit power needed is relatively high (some 10 W) which causes problems for battery powered devices. These satellites cannot be used for small mobile phones. The biggest problem for voice and data communication is the high latency of over 0.25 s one-way many retransmission schemes which are known from fixed networks fail. Due to the large footprint, either frequencies cannot be reused, or the GEO satellite needs special antennas focusing on a smaller footprint. Transferring a GEO into orbit is very expensive.

MEDIUM EARTH ORBIT (MEO)

MEOS operate at a distance of about 5,000-12,000 km. Up to now there have not been many satellites in this class, but some upcoming systems (e.g., ICO) use this class for various reasons.

MEOS can be positioned somewhere between LEOS and GEOS, both in terms of their orbit and due to their advantages and disadvantages.




GEO: Geostationary Earth Orbit satellite
35,800km height
MEO: Medium Earth Orbit satellite
5,000km to 12,000km
LEO: Low Earth Orbit satellite
500km to 1,500km

ADVANTAGES OF MEO

Using orbits around 10,000 km, the system only requires a dozen satellites which is more than a GEO system, but much less than a LEO system. These satellites move more slowly relative to the earth's rotation allowing a simpler system design (satellite periods are about six hours).

Depending on the inclination, a MEO can cover larger populations, so requiring fewer handovers.

DISADVANTAGES OF MEO

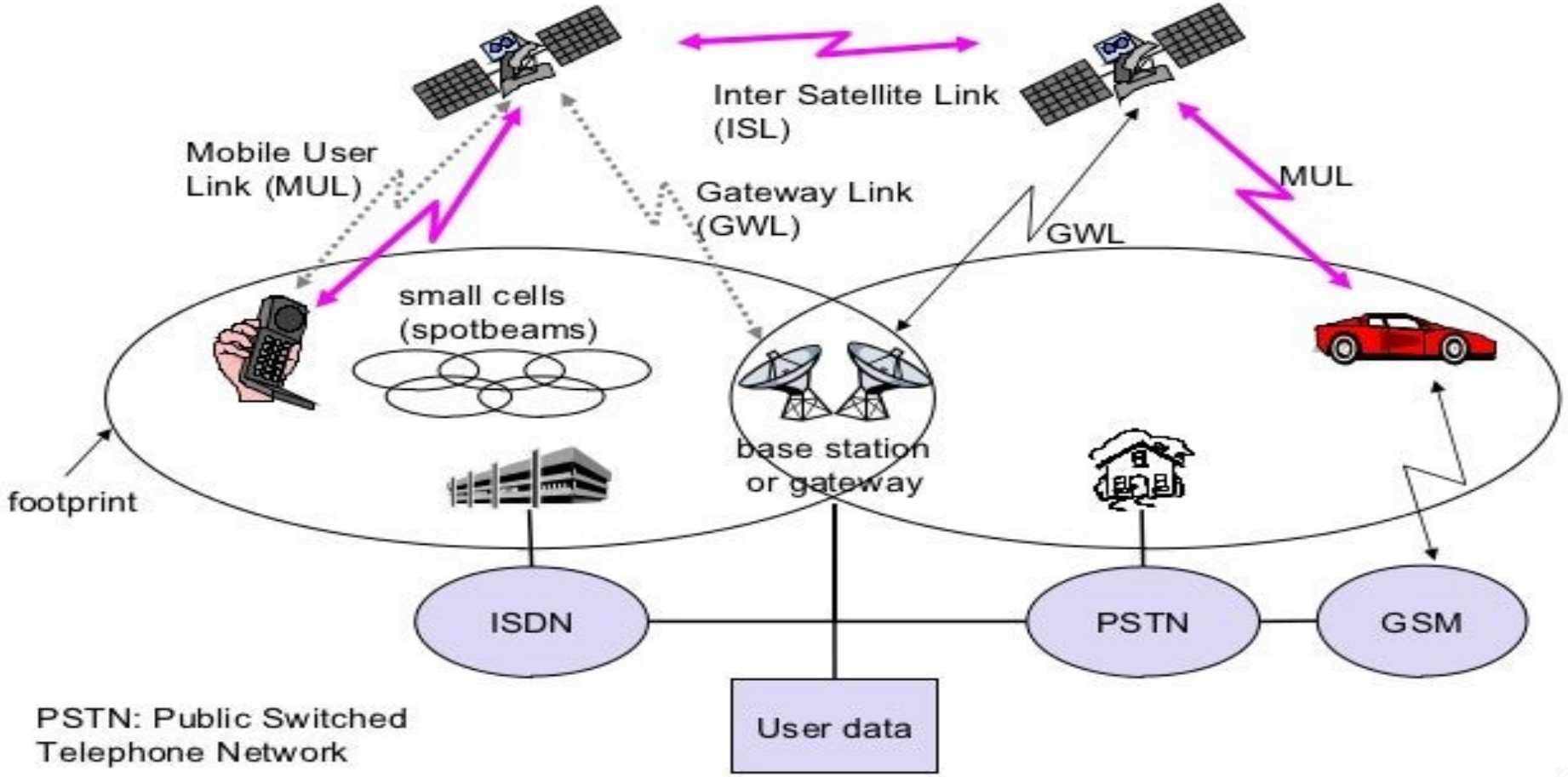
Again, due to the larger distance to the earth, delay increases to about 70-80ms. The satellites need higher transmit power and special antennas for smaller

LOW EARTH ORBIT (LEO)

As LEOs circulate on a lower orbit, it is obvious that they exhibit a much shorter period (the typical duration of LEO periods are 95 to 120 minutes), Additionally, L.EO systems try to censure a high elevation for every spot on earth to provide a high-quality communication link. Each LEO satellite will only be visible from the earth tor around ten minutes. A further classification of LEOs into little LEOs with low bandwidth services (some 100 bit/s), big LEOs (some 1,000 bit/s) and broadband LEOs with plans reaching into the Mbit/s range can be found in Comparetto (1997).

While some time ago LEO satellites were mainly used for espionage, several of the new satellite systems now rely on this class using altitudes of 500-1,500 km

Classical satellite systems



PSTN: Public Switched
Telephone Network

ADVANTAGES OF LEO

Using advanced compression schemes, transmission rates of about 2,1O0 bit/s can be enough for voice communication. LEOs even provide this bandwidth for mobile terminals with omni-directional antennas using low transmit power in the range of 1W. The delay for packets delivered via a LEO Is relatively low (approx 10 ms). The delay Is comparable to long-distance wired connections (about 5-10 ms). Smaller footprints of LEOS allow for better frequency reuse, similar to the concepts used for cellular networks (Gavish, 1998). LEOs can provide a much higher elevation in polar regions and so better global coverage.

DISADVANTAGES OF LEO

The biggest problem of the LEO concept Is the need for many satellites if global coverage is to be reached. Several concepts involve 50-200 or even more satellites in orbit. The short time of visibility with a high elevation requires additional mechanisms for connection handover between different satellites. The high number of satellites combined with the fast movements results in a high complexity of the whole satellite system.

One general problem of LEOs is the short lifetime of about five to eight years due to atmospheric drag and radiation from the inner Van Allen belt. Assuming 48 satellites and a lifetime of eight years (as expected for the system Globalstar), a new satellite would be needed every two months. The low latency via a single LEO is only half of the story. Other factors are the need for routing of data packets from satellite to satellite (or several times from base stations to satellites and back) if a user wants to communicate around the world. Due to the large footprint, a GEO typically does not need this type of routing, as senders and receivers are most likely in the same footprint.

HIGHLY ELLIPTICAL ORBIT (HEO):

This class comprises all satellites with non-circular orbits. Currently, only a few commercial communication systems using satellites with elliptical orbits are planned. These systems have their perigee over large cities to improve communication quality. The Van Allen radiation belts, be heights of about 2,000-6,000 km (inner Van Allen belt) and about 15,000-30,000 km (outer Van Allen belt) respectively make satellite communication very difficult in these orbits.

ROUTING

A satellite system together with gateways and fixed terrestrial networks as has to route data transmissions from one user to another as any other network does. Routing in the fixed segment (on earth) is achieved as usual, while two different solutions exist for the satellite network in space. If satellites offer ISLs, traffic can be routed between the satellites. If not, all traffic is relayed to earth, routed there, and relayed back to a satellite. Assume two users of a satellite network exchange data. If the satellite system supports ISLs, one user sends data up to a satellite and the satellite forwards it to the one responsible for the receiver via other satellites. This last satellite now sends the data down to the earth. This means that only one uplink and one downlink per direction is needed. The ability of routing within the satellite network reduces the number of gateways needed on earth. If a satellite system does not offer 1SLs, the user also sends data up to a satellite, but now this satellite forwards the data to a gateway on earth. Routing takes place in fixed networks as usual until another gateway is reached which is responsible for the satellite above the receiver. Again data is sent up to the satellite which forwards it down to the receiver. This solution requires two uplink and two downlinks. Depending on the orbit and the speed of routing in the satellite network compared to the terrestrial network, the solution with ISLs might offer lower latency. The drawbacks of ISLs are higher system complexity due to additional antennas and routing hard- and software for the satellites.

LOCALIZATION

Localization of users in satellite networks is similar to that of terrestrial cellular networks. One additional problem arises from the fact that now the base stations', i.e., the satellites, move as well. The gateways of a satellite network maintain several registers. A **home location register** (HLR) stores all static information about a user as well as his or her current location. The last known location of a mobile user is stored in the **visitor location register** (VLR). Functions of the VLR and HLR are similar to those of the registers in, e.g., GSM. A particularly important register in satellite networks is the **satellite user mapping registe**r (SUMR). This stores the current position of satellites and a mapping of each user to the current satellite through which communication with a user is possible.

Registration of a mobile station is achieved as follows. The mobile station initially sends a signal which one or several satellites can receive. Satellites receiving such a signal report this event to a gateway. The gateway can now determine the location of the user via the location of the satellites. User data is requested from the user's HLR, VLR and SUMR are updated. Calling a mobile station is again similar to GSM. The call is forwarded to a gateway which localizes the mobile station using HLR and VLR. With the help of the SUMR, the appropriate satellite for communication can be found and the connection can be set up.

HANDOVER

An important topic in satellite systems using MEOs and in particular LEOS is handover. Imagine a cellular mobile phone network with fast moving base stations. This is exactly what such satellite systems are - each satellite represents a base station for a mobile phone. Compared to terrestrial mobile phone networks, additional instances of handover can be necessary due to the movement of the satellites.

- **Intra-satellite handover**: A user might move from one spot beam of a satellite to another spot beam of the same satellite. Using special antennas, a satellite can create several spot beams within its footprint. The same effect might be caused by the movement of the satellite.
- **Inter-satellite handover**: If a user leaves the footprint of a satellite or if the satellite moves away, a handover to the next satellite takes place. This might be a hard handover switching at one moment or a soft hand overusing both satellites (or even more) at the same time (as this is possible with CDMA systems). Inter-satellite handover can also take place between satellites if they support ISLs. The satellite system can trade high transmission quality for handover frequency. The higher the transmission quality should be, the higher the elevation angles that are needed. High elevation angles imply frequent handovers which in turn, make the system more complex.

- **Gateway handover**: While the mobile user and satellite might still have good contact, the satellite might move away from the current gateway. The satellite has to connect to another gateway.
- **Inter-system handover**: While the three types of handover mentioned above take place within the satellite-based communication system, this type of handover concerns different systems. Typically, satellite systems are used in remote areas if no other network is available. As soon as traditional cellular networks are available, users might switch to this type usually because it is cheaper and offers lower latency. Current systems allow for the use of dual-mode (or even more) mobile phones but unfortunately, seamless handover between satellite systems and terrestrial systems or vice versa has not been possible up to now.

HANDOVER CLASSIFICATIONS

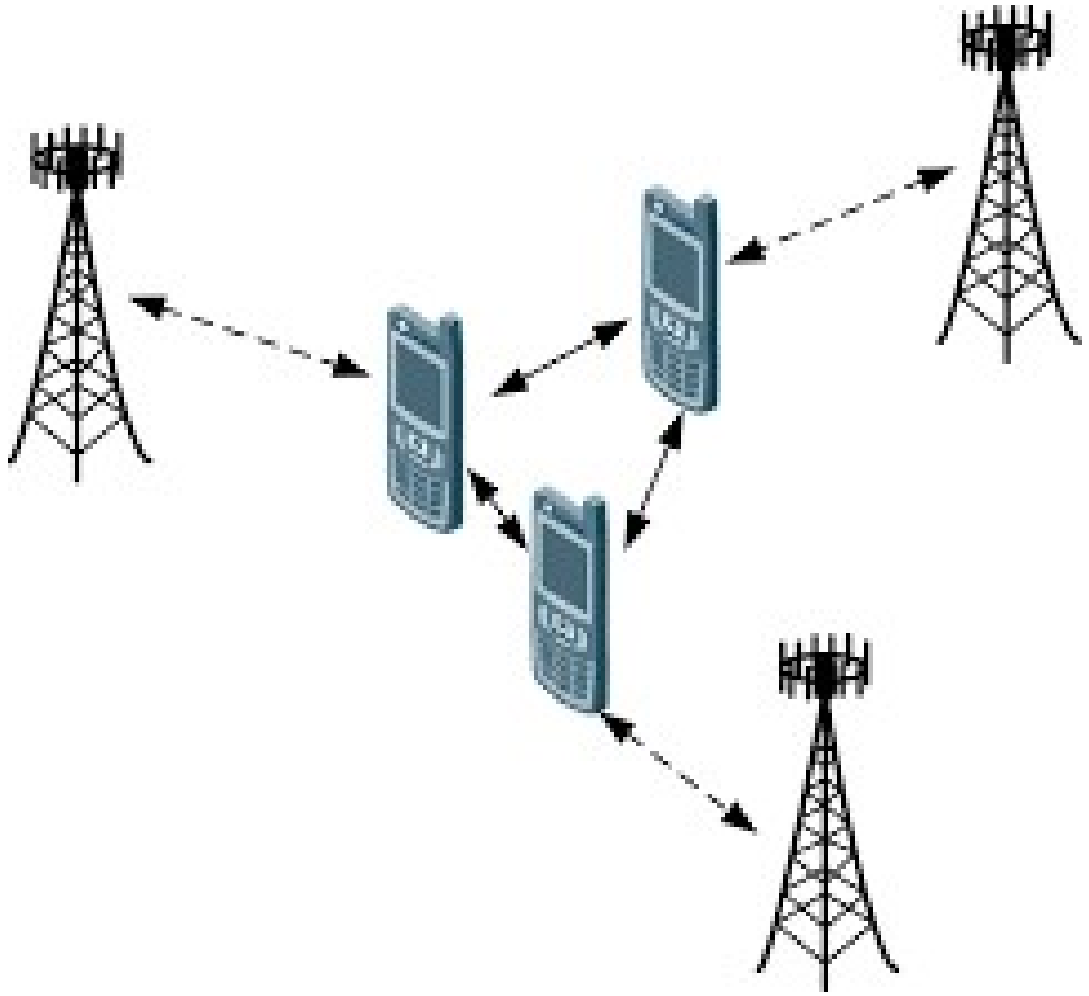
UMTS knows two basic classes of handovers:

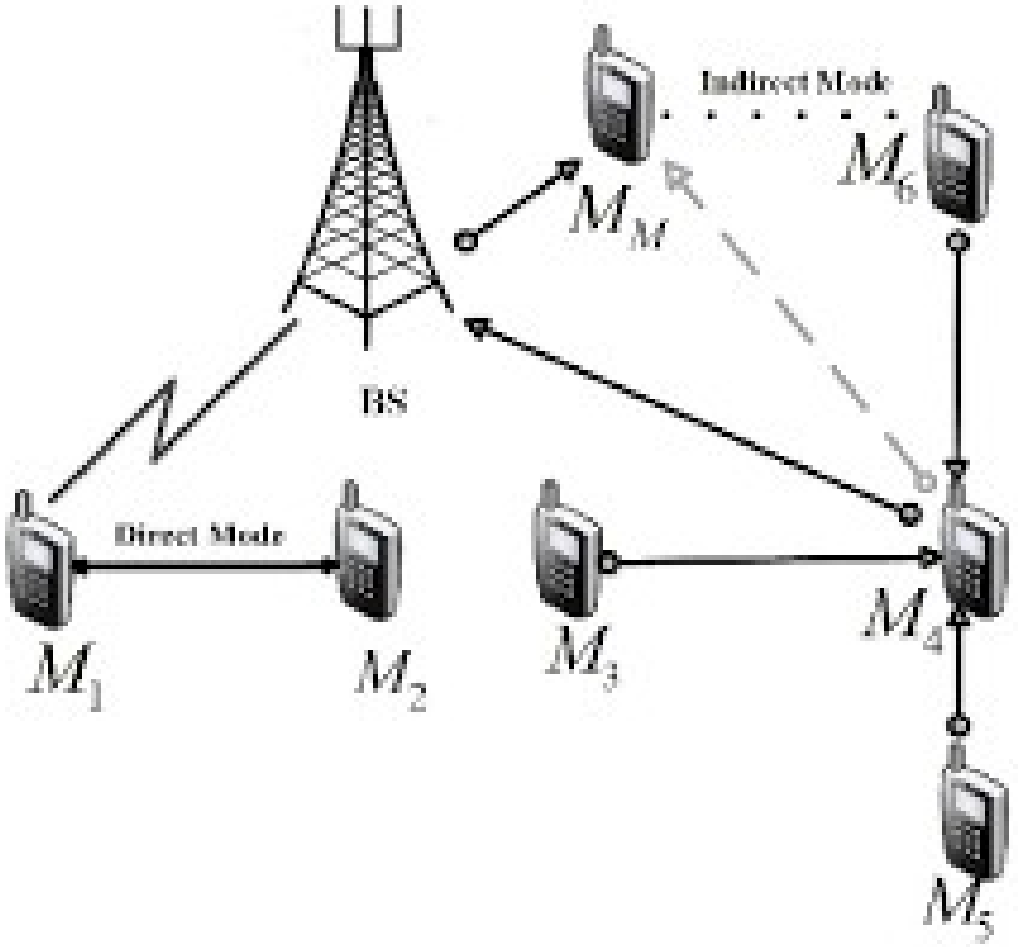
•**Hard handover**: This handover type is already known from GSM and other TDMA/FDMA systems. Switching between different antennas or different systems is performed at a certain point in time. UTRA TDD can only use this type. Switching between TDD cells is done between the slots of different frames. **Inter frequency handover**, i.e., changing the carrier frequency, is a hard handover. Receiving data at different frequencies at the same time requires a more complex receiver compared to receiving data from different sources at the same carrier frequency. Typically, all inter system handovers are hard handovers in UMTS. This includes handovers to and from GSM or other IMT-2000 systems. A special type of handover is the handover to a satellite system (inter-segment handover), which is also a hard handover, as different frequencies are used. However, it is unclear what technology will be used for satellite links if it will ever come. To enable a UE to listen into GSM or other frequency bands, UMTS specifies a compressed mode transmission for UTRA FDD. During this mode a UE stops all transmission. To avoid data loss, either the spreading factor can be lowered before and after the break in transmission (i.e., more data can be sent in shorter time) or lessdata is sent using different coding

- **Soft handover**: This is the real new mechanism in UMTS compared to GSM and is only available in the FDD mode. Soft handovers are well known from traditional CDMA networks as they use macro diversity, a basic property of CDMA. A UE can receive signals from up to three different antennas, which may belong to different node Bs. Towards the UE the RNC splits the data stream and forwards it to the node Bs. The UE combines the received data again. In the other direction, the UE simply sends its data which is then received by all node Bs involved. The RNC combines the data streams received from the node Bs. The fact that a UE receives data from different antennas at the same time makes a handover soft. Moving from one cell to another is a smooth, not an abrupt process.

LOCALIZATION AND CALLING

One fundamental feature of the GSM system is the automatic, worldwide localization of users. The system always knows where a user currently is, and the same phone number is valid worldwide. To provide this service, GSM performs periodic location updates even if a user does not use the mobile station (provided that the MS is still logged into the GSM network and is not completely switched off). The HLR always contains information about the current location (only the location area, not the precise geographical location), and the VLR currently responsible for the MS informs the HLR about location changes. As soon as an MS moves into the range of a new VLR (a new location area), the HLR sends all user data needed to the new VLR. Changing VLRs with uninterrupted availability of all services is also called roaming. Roaming can take place within the network of one provider, between two providers in one country (national roaming is, often not supported due to competition between operators), but also between different providers in different countries (international roaming). Typically, people associate international roaming with the term roaming as it is this type of roaming that makes GSM very attractive: one device, over 190 countries! To locate an MS and to address the MS, several numbers are needed:





ig. 1 A M2M communication system over cellular network

- **Mobile station international ISDN number (MSISDN):** The only important number for a user of GSM is the phone number. Remember that the phone number is not associated with a certain device but with the SIM, which is personalized for a user. The MSISDN follows the ITU-T standard E.164 for addresses as it is also used in fixed ISDN networks. This number consists of the country code (CC) (e.g., +49 179 1234567 with 49 for Germany), the national destination code (NDC) (1.e., the address of the network provider, e.g, 179), and the subscriber number (SN)

- **International mobile subscriber identity (IMS):** GSM uses the IMSI for internal unique identification of a subscriber. IMSI consists of a mobile country code (MCC) (e.g., 240 for Sweden, 208 for France), the **mobile network code (MNC)** (1.e., the code of the network provider), and finally the **mobile subscriber identification number (MSIN).**

- **Temporary mobile subscriber identity (TMS):** To hide the IMSI, which would give away the exact identity of the user signaling over the air interface, GSM uses the 4 byte TMSI for local subscriber identification. TMSI is selected by the current VLR and is only valid temporarily and within the location area of the VLR (for an ongoing communication TMSI and LAl are sufficient to identify a user; the IMSI is not needed). Additionally, a VLR may change the TMSI periodically.

- **Mobile station' roaming number (MSRN):** Another temporary address that hides the identity and location of a subscriber is MSRN. The VLR generates this address on request from the MSC, and the address is also stored in the HLR. MSRN contains the current **visitor country code (VCC),** the visitor national destination code (VNDC), the identification of the current MSC together with the subscriber number. The MSRN helps the HLR to find a subscriber for an incoming call.

Cellular systems require handover procedures, as single cells do not cover the whole service area, but, e.g., only up to 35 km around each antenna on the countryside and some hundred meters in cities (Tripathi, 1998). The smaller the cell size and the faster the movement of a mobile station through the cells (up to 250 km/h for GSM), the more handovers of ongoing calls are required. However, a handover should not cause a cut-off, also called call drop. GSM aims at maximum handover duration of 60 ms. There are two basic reasons for a handover (about 40 have been identified in the standard):

- The mobile station moves out of the range of a BTS or a certain antenna of a BTS respectively. The received signal level decreases continuously until it falls below the minimal requirements for communication. The error rate may grow due to interference, the distance to the BTS may be too high (max. 35 km) etc. - all these effects may diminish the quality of the radio link and make radio transmission impossible in the near future.
- The wired infrastructure (MSC, BSC) may decide that the traffic in one cell is too high and shift some MS to other cells with a lower load (if possible). Handover may be due to load balancing

- **Intra-cell handover**: Within a cell, narrow-band interference could make transmission at a certain frequency impossible, The BSC could then decide to change the carrier frequency (scenario 1).
- **Inter-cell, intra-BSC handover**: This is a typical handover scenario. The mobile station moves from one cell to another but stays within the control of the same BSC. The BSC then performs a handover, assigns a new radio channel in the new cell and releases the old one (scenario 2).
- **Inter-BSC, intra-MSC handover**: As a BSC only controls a limited number of cells; GSM also has to perform handovers between cells controlled by different BSCs. This handover then has to be controlled by the MSC (scenario 3).
- **Inter MSC handover**: A handover could be required between two cells

belonging to different MSCs. Now both MSCs perform the handover

together (scenario 4).

Data communications

Data communications involves the exchange of digital information between devices over networks or transmission media. This process enables everything from email and video calls to cloud computing and IoT applications.

Key Components: Core elements include a sender device, a receiver, the message (data like text or video), and a transmission medium such as cables or wireless signals. Protocols govern sequencing, routing, and flow control to ensure reliable delivery without loss or overload.

Network Types

Local Area Networks (LANs) connect devices in small areas like offices. Local Area Networks (LANs) connect devices like computers, printers, and servers within a limited geographic area, such as a home, office, or building, typically up to a few kilometers. They enable high-speed resource sharing and communication using technologies like Ethernet or Wi-Fi.

Data communication

Key Characteristics

LANs provide fast data rates from 100 Mbps to 10 Gbps, low latency due to short distances, and private ownership for secure, cost-effective setups. They support centralized management and easy security protocols to control access.

Common Components

Essential hardware includes network switches for traffic direction, routers for internet connectivity, Wi-Fi access points for wireless devices, and Ethernet cables or NICs in each device. Software like TCP/IP handles addressing and protocols.

Topologies and Types

Popular topologies are star (central switch), bus (linear cable), and ring (looped connections). Types include wired Ethernet LANs and wireless WLANs, with client-server models for larger setups.

Topologies and Types

Metropolitan Area Networks (MANs) span cities.

Metropolitan Area Networks (MANs) connect multiple Local Area Networks (LANs) across a city or large campus, typically spanning 5 to 50 kilometers. They bridge the gap between small-scale LANs and expansive Wide Area Networks (WANs).

Key Characteristics

MANs deliver high-speed data transfer using fiber optic cables or wireless links, offering scalability, reliability, and broad coverage for urban environments. They often combine public and private infrastructure for efficient interconnectivity among offices, campuses, or city services.

Common Applications

- Organizations use MANs for linking branch offices, universities, or government facilities within a city, supporting video conferencing, resource sharing, and high-bandwidth services like cable TV distribution. Telecom providers leverage them for urban internet backbones.

Unlike LANs (building-limited) or WANs (global), MANs focus on metropolitan scales with dedicated backbones for faster, more controlled traffic than internet-based WANs. They reduce latency for city-wide operations compared to broader networks.

Wide Area Networks (WANs) link distant locations using satellites or fiber optics.

Wide Area Networks (WANs) connect multiple Local Area Networks (LANs) across large geographical areas like cities, countries, or continents, enabling global data sharing and communication. They form the backbone of the internet and corporate connectivity.

Key Characteristics

- WANs span vast distances using technologies like leased lines, fiber optics, satellites, and MPLS for high-speed, reliable transfer despite higher latency than LANs. They mix public infrastructure (internet) with private links for security and efficiency.

Switched WANs share resources via central hubs like cellular towers or ATM for dynamic traffic. Point-to-point WANs provide dedicated leased lines between sites for exclusive, high-bandwidth access

Transmission Modes

Data moves in simplex (one-way), half-duplex (alternating two-way), or full-duplex (simultaneous two-way) modes. Channels include twisted pair, coaxial cable, fiber optics, or radio waves, with bandwidth determining transfer speed.

Modern Applications

- Advancements like 5G support self-driving cars, remote surgery, and augmented reality. These networks underpin business efficiency, instant global connectivity, and digital services from online banking to streaming. A computer network interconnects computing devices like computers, servers, and printers to enable resource sharing, data exchange, and communication over wired or wireless links. This setup supports everything from local file sharing to global internet access.

Hardware Components

Essential hardware includes hosts (end devices such as PCs and mobiles), connecting devices like routers (for inter-network routing), switches (for efficient local data directing), hubs (basic multi-device connectors), modems (signal conversion for external lines), and network interface cards (NICs) in each device. Transmission media like cables or wireless signals link these elements.

Software Elements

Protocols such as TCP/IP and applications form the software layer, handling data formatting, error correction, addressing, and services like web browsing or email. Operating systems manage network operations alongside these protocols.

Network Types

- Common scales include LANs for buildings, WANs for regions, MANs for cities, and PANs for personal spaces, with topologies like star (hub-centered) or mesh (fully interconnected).

Computer network.

Computer networks classify primarily by geographical scale, connecting devices for data sharing and communication over varying distances. Common types include PANs for personal use, LANs for local setups, MANs for urban areas, and WANs for wide-reaching connections.

Personal Area Network (PAN)

PANs cover small ranges like 10 meters, linking personal devices such as smartphones, laptops, and wearables via Bluetooth or USB. They enable short-range tasks like file transfer between gadgets.

Local Area Network (LAN)

LANs span buildings or campuses up to a few kilometers, using Ethernet or Wi-Fi for high-speed connections among computers and printers. Offices and homes rely on LANs for resource sharing.

Metropolitan and Wide Area Networks

- MANs connect multiple LANs across a city (10-50 km) for services like cable TV. WANs, the largest type, link global sites via satellites or fiber, powering the internet for worldwide access.

Other Classifications

Networks also divide by topology (star, mesh, bus), ownership (private, public), or design (peer-to-peer vs. client-server). Specialized types include WLANs (wireless LANs) and VPNs for secure remote access

RADIO WAVE PROPAGATION

- Radio wave propagation describes how electromagnetic waves travel from transmitter to receiver through modes like ground wave (follows Earth's surface for low frequencies), sky wave (ionospheric reflection for long-distance HF), and space wave (line-of-sight with direct and tropospheric paths). Transmission lines guide RF signals efficiently from sources to antennas, minimizing losses via structures like coaxial cables or waveguides.

Propagation Modes

Ground wave propagation suits AM radio over hundreds of kilometers by hugging terrain, affected by soil conductivity. Sky wave enables global reach via ionosphere bounces but varies with solar activity. Space wave supports VHF/UHF line-of-sight up to 50-100 km, enhanced by tropospheric ducting for medium ranges.

Transmission Line Fundamentals

Transmission lines carry RF power with controlled impedance to prevent reflections, using types like bifilar (parallel wires), coaxial (shielded for low loss), or waveguides (hollow metal for microwaves). Key parameters include characteristic impedance, attenuation, and velocity factor, critical above 1 MHz where cable lengths act as lines.

Applications in Communications

These concepts underpin wireless systems: propagation modes determine coverage in mobile networks, while transmission lines connect transmitters to antennas in radar or 5G base stations. Factors like frequency, terrain, and weather influence performance.